MGM Macau and Cotai

Resource Efficiency Management Report

December 2022

Energenz Consulting Limited

EXECUTIVE SUMMARY

Energenz has been working closely with MGM to implement the EMOs as well as identify additional opportunities to achieve ISO certification energy targets. Energenz appreciates the great support and trust from MGM team over the years, in particular the Facility Management and the Sustainability Teams.

Key Performance Indicators

Energenz reports both normalized and actual savings on a monthly basis for both properties. The normalized savings are determined based on a 2019 baseline, with aligned annual savings targets to meet both ISO50001 as well as Energenz's retainer contract requirements. Please note that the reported normalized savings achieved are to-date and cumulative for each month of the respective year.

Table 1. ISO and Retainer Contract Energy Performance Targets for 2025

MGM Properties	Baseline	Energy Savings	2021	Normalized Savings Achieved Up to December 2022	2023	2024	2025
Macau	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-5.9%**	**-10.8%**	-	-	**-**
Cotai	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-6.4%**	**-16.9%**	-	-	**-**

**MGM Cotai ISO energy performance: e-bus consumption is included.*

The actual monthly savings are determined against the 2019 baseline consumption and cumulative for each month of 2022.

Table 2. Internal Actual Energy Performance Targets for 2025

MGM Properties	Baseline	Actual Saving Target[1]	Savings Achieved Up to December 2022
Macau	2019	15%	**-8.3%**
Cotai	2019	15%	**-15.4%**

[1] Actual Energy Savings Target for 2025 has been established based on future CAPEX projects, following review with MGM teams.

Highlighted Changes in Operation

Macau

1. MGM Macau has resumed normal operation on 23 July 2022, all of the previous low occupancy energy saving measures have been deactivated.
2. Hotel tower L19 PAU dehumidification mode has been switched off on 8 August 2022 to reduce LPG boiler usage.
3. The chiller CHWST setpoint was adjusted to 8.5 - 9.5°C on 11 October 2022 for energy saving. Due to the L2 & L2M server room high temperature alarm, it was fine-tuned as 8.5 - 9.0°C on 24 October 2022.
4. Main LPG Boiler has been switched ON from 30 November 2022 until end of the year as the outdoor temperature has dropped significantly during December 2022.

Cotai

1. MGM Cotai has resumed normal operation on 23 July 2022, all of the previous low occupancy energy saving measures have been deactivated.
2. Installation of 10 units of variable speed drives (VSD) for hotel tower exhaust fans (EF) have been completed by 22 August 2022. The new VSDs have been set at 45 Hz for energy saving and enhancing the positive differential air pressure into hotel towers staircases.
3. The low voltage switchboard (no.18 200A 4P MCCB of MDB-NA01B) for MGM Cotai MDM restaurant kitchen was switched off from 22 September 2022 for power enhancement work. The MDM restaurant kitchen is temporarily being supplied by another power source from SMDB-XA01B-FUT (Meter number XAB1B-16).
4. Guo Pei shop has installed 3 dehumidifiers on 17 October 2022 for space dehumidification. It is estimated additional 4,500 kWh/month electricity will be consumed.
5. NG Boiler has been switched ON in December 2022 for supporting space heating and domestic hot water use during cold weather.

Consumption Intensity

The YTD electricity and water use intensities are tracked on a monthly basis for both properties:

Table 3. Year to Date Actual Electricity Use Intensity (EUI)

MGM Properties	YTD EUI (kWh/m^2)	% Change Compared to 2021
Macau	**24.5**	**-5.1%**
Cotai	**24.0**	**-9.9%**

Table 4. Year to Date Actual Water Use Intensity (WUI)

MGM Properties	YTD WUI (m^3/ m^2)	% Change Compared to 2021
Macau	**0.19**	**-8.4%**
Cotai	**0.13**	**-21.6%**

Note: Cotai Area: 374,138 m^2 Macau Area: 211,545 m^2

ISO 50001 Certification Audit

Macau

The HKQAA 2022 surveillance audit was completed on 30 March 2022. There is no nonconformity (NC) and areas for improvement (AFI) from auditors.

Cotai

The HKQAA 2022 recertification audit was completed on 30 September 2022. There is no nonconformity (NC) and areas for improvement (AFI) from auditors.

Target Milestones Completed in Year 2022

The following activities were completed in year 2022:

- Performed ISO 50001 2022 Internal and External Audit for MGM Macau and MGM Cotai
- Deployed and fine-tuned the DGLux dashboard with heatmap development extension
- Performed 2021 Carbon and Waste Audit for MGM Macau and MGM Cotai
- Assisted MGM with the COTAI phase 2 planning
- Investigated and implemented the below list of projects during the Year 2022

Table 5. MGM Macau Year 2022 Projects

Item	Initiatives	Status
Electricity & LPG		
1	Chilled Water Pump Retrofit (Phase 1)	Postpone to 2023
2	PICCV Implementation (Phase 3)	Completed
3	Air Side Equipment IE4 Motor Replacement	Completed
4	Neon Light LED Replacement (Phase 2)	Target to complete by early 2023
5	Replacement of L2M Steam-to-Water Heat Exchanger to Fully Utilize Heat Pump	Completed

Table 6. MGM Cotai Year 2022 Projects

Item	Initiatives	Status
Electricity & NG		
1	Chiller Plant Control Optimization (Chiller/CHWP/CDWP/CT)	On-going
2	Hot Water System Upgrade 1. Increase Max. No of Plate for Pre-heat Compact Unit 2. Upgrade Booster Compact Unit 3. Install Electric Heater for Booster Calorifier	On-going
3	AHU Static Pressure & Supply Air Temperature Set-Point Reset (Phase 2)	Postpone to 2023
4	PICCV Implementation (Phase 3)	Completed
5	Hotel Towers Building Pressurization and EAF Optimization	Completed
6	Kitchen Exhaust Fan (KEF) Steps Speed Control Optimization	Completed
7	Carpark Ventilation Fans Control Optimization with IoT Sensor Installation (Phase 2)	Completed
8	Centralized RO Pump Set Replacement (10 Units)	Target to complete by mid 2023
Water		
9	Extension of CDP Water Recycling (Phase 3 & 4)	Completed
10	Rainwater Harvesting for CT Make Up (Phase 1)	Completed
11	Non-potable Centralized RO Optimization for Cooling Tower	Completed

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ACKNOWLEDGEMENT

Energenz has been working with MGM Team as the energy manager to achieve energy savings in MGM properties and feed operational intelligence into the design.

The major goal of this project is not only to realize operation energy savings but also to extend the life of equipment, hence reducing CAPEX. MGM has demonstrated continuous commitment to be an environmentally and socially responsible business in the past years. In addition to that Energenz has assisted MGM Macau and Cotai to achieve several other energy management and sustainability awards and certifications, such as ISO 50001, CEM Energy Savings Award, Macao Green Hotel Award and China 3 Star Green Building Operation Label.

Energenz appreciates the great support and trust from MGM Team over the years, especially the Facilities Management and the Sustainability Teams.

1. Portfolio Performance Summary

1.1 Electricity Use Intensity (EUI)

Macau's electricity use intensity (EUI) in December 2022 had decreased by **7.3%** compared to the same month in 2021 and decreased by **5.1%** in 2022 compared to 2021. Meanwhile, Cotai's EUI in December 2022 has decreased by **8.8%** compared to the same month in 2021 and decreased by **9.9%** in 2022 compared to 2021.

Table 7. Actual Electricity Consumption Intensity (EUI) Comparison

Monthly Period	Macau			Cotai		
	Year 2021 Actual EUI (kWh/m²)	Year 2022 Actual EUI (kWh/m²)	% Change	Year 2021 Actual EUI (kWh/m²)	Year 2022 Actual EUI (kWh/m²)	% Change
January	20.4	22.6	+11.0%	21.8	22.3	+2.1%
February	19.8	20.2	+1.8%	20.4	20.1	-1.4%
March	23.3	24.4	+4.5%	24.6	23.6	-4.0%
April	23.8	24.3	+1.8%	25.4	24.0	-5.6%
May	28.9	25.9	-10.5%	30.6	25.6	-16.3%
June	28.7	27.3	-4.9%	29.7	27.0	-9.3%
July	30.8	25.9	-15.8%	31.7	24.3	-23.4%
August	29.8	28.1	-5.8%	30.5	26.3	-13.7%
September	30.3	26.8	-11.8%	30.0	25.6	-14.6%
October	26.8	24.6	-8.3%	27.3	24.4	-10.6%
November	23.9	22.7	-4.9%	24.0	23.3	-3.0%
December	22.9	21.2	-7.3%	23.1	21.0	-8.8%
Average	**25.8**	**24.5**	**-5.1%**	**26.6**	**24.0**	**-9.9%**

1.2 Water Use Intensity (WUI)

Macau's water use intensity (WUI) in December 2022 decreased by **13.4%** compared to the same month in 2021 and decreased by **8.4%** in 2022 compared to 2021. Meanwhile, Cotai's WUI in December 2022 decreased by **29.7%** compared to the same month in 2021 and decreased by **21.6%** for 2022 compared to 2021.

Table 8. Actual Water Consumption Intensity (WUI) Comparison

Monthly Period	Macau			Cotai		
	Year 2021 Actual WUI (m^3/m^2)	Year 2022 Actual WUI (m^3/m^2)	% Change	Year 2021 Actual WUI (m^3/m^2)	Year 2022 Actual WUI (m^3/m^2)	% Change
January	0.19	0.18	-7.2%	0.116	0.119	+3.2%
February	0.19	0.20	+1.1%	0.14	0.12	-10.9%
March	0.20	0.19	-7.4%	0.15	0.12	-19.3%
April	0.24	0.21	-9.1%	0.18	0.14	-21.2%
May	0.25	0.23	-7.7%	0.21	0.15	-28.2%
June	0.21	0.19	-9.3%	0.18	0.12	-30.7%
July	0.22	0.15	-32.6%	0.19	0.10	-46.2%
August	0.20	0.19	-3.8%	0.17	0.13	-21.8%
September	0.23	0.22	-3.7%	0.19	0.15	-20.9%
October	0.18	0.19	+5.6%	0.15	0.14	-7.6%
November	0.21	0.19	-10.2%	0.15	0.14	-8.5%
December	0.19	0.17	-13.4%	0.16	0.11	-29.7%
Average	**0.21**	**0.19**	**-8.4%**	**0.16**	**0.13**	**-21.6%**

2. MGM Macau Performance

2.1 Electricity Consumption

The actual electricity consumption and cost in December 2022 decreased by **7.3%** and **4.2%** respectively compared to the same month in 2021. To date, MGM Macau has decreased the actual electricity consumption by **5.1%** and associated cost decreased by **1.3%** respectively.

Table 9. Macau - Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2021	2022	% Change - Actual	2021	2022	% Change - Actual Cost	
January	4,310,160	4,784,280	+11.0%	4,717,260	5,519,350	+17.0%	+15.8%
February	4,190,760	4,266,240	+1.8%	4,591,170	4,942,590	+7.7%	-90.8%
March	4,931,880	5,152,320	+4.5%	5,387,480	5,938,160	+10.2%	-6.9%
April	5,042,160	5,131,680	+1.8%	5,808,340	6,183,420	+6.5%	-11.4%
May	6,122,640	5,478,840	-10.5%	7,034,530	6,595,030	-6.2%	-32.6%
June	6,072,960	5,777,040	-4.9%	7,681,070	7,567,480	-1.5%	-1.0%
July	6,506,760	5,480,880	-15.8%	8,217,570	7,172,780	-12.7%	+4.1%
August	6,306,840	5,938,920	-5.8%	7,960,170	7,799,020	-2.0%	+1.7%
September	6,418,920	5,659,200	-11.8%	8,118,840	7,450,220	-8.2%	-0.1%
October	5,665,560	5,194,440	-8.3%	6,517,390	6,209,630	-4.7%	+7.9%
November	5,058,840	4,810,800	-4.9%	5,835,330	5,737,570	-1.7%	+14.7%
December	4,849,920	4,494,480	-7.3%	5,592,970	5,360,740	-4.2%	-93.6%
Total/Average	**65,477,400**	**62,169,120**	**-5.1%**	**77,462,120**	**76,475,990**	**-1.3%**	**-7.5%**

Important note: *Cooling Degree Days (CDDs) is the main variable taken into consideration on electricity consumption normalization.*

Energenz summarized the key items which could result in electricity consumption decrease below:

- Dec 2022 chiller plant electricity consumption has decreased by 21.5% compared to the same month in 2021 due to significant drop of CDD.

Table 10 presents the comparison of electricity rates in the years 2021 and 2022. The electricity rate in December 2022 increased by **3.4%** compared to the same month in 2021.

Table 10. Macau - Summary of Actual Monthly Electricity Rate

Monthly Period	Year 2021 – Electricity Rate (MOP/kWh)	Year 2022 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.09	1.15	+5.4%
February	1.10	1.16	+5.7%
March	1.09	1.15	+5.5%
April	1.15	1.20	+4.6%
May	1.15	1.20	+4.8%
June	1.26	1.31	+3.6%
July	1.26	1.31	+3.6%
August	1.26	1.31	+4.0%
September	1.26	1.32	+4.1%
October	1.15	1.20	+3.9%
November	1.15	1.19	+3.4%
December	1.15	1.19	+3.4%
Total/Average	**1.17**	**1.23**	**+4.3%**

Table 11 summarizes the monthly normalized electricity consumption. The actual electricity consumption in December 2022 decreased by **2.5%** compared to 2019 normalized consumption in the same month.

Table 11. Macau - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2022 Actual Electricity Consumption (kWh)	% Change - Normalized
January	4,790,450	4,784,280	-0.1%
February	4,631,715	4,266,240	-7.9%
March	5,342,712	5,152,320	-3.6%
April	5,631,086	5,131,680	-8.9%
May	5,930,696	5,478,840	-7.6%
June	6,468,840	5,777,040	-10.7%
July	6,859,773	5,480,880	-20.1%
August	6,628,440	5,938,920	-10.4%
September	6,732,439	5,659,200	-15.9%
October	6,115,935	5,194,440	-15.1%
November	5,576,061	4,810,800	-13.7%
December	4,611,549	4,494,480	-2.5%
Total/Average	**69,319,697**	**62,169,120**	**-10.3%**

2.2 LPG Consumption

LPG consumption can be broken down into three (3) categories: boiler, hotel upper zone DHW, and kitchens/restaurants. Since the boiler and restaurants LPG utilization cover over 95% of the overall site LPG consumption, Energenz focuses the analysis on those two areas.

According to Table 12, the actual LPG consumption in December 2022 increased by **14.7%** and cost decreased **11.2%** respectively compared to the same month in 2021. To date, MGM Macau has increased the actual LPG consumption by **5.6%** and associated costs increased by **18.9%**

Table 12. Macau - Summary of Actual Monthly LPG Consumption and Cost

Monthly Period	Actual LPG Consumption (kg)			Actual LPG Cost (MOP)			% Change – Min OAT
	2021	2022	% Change - Actual	2021	2022	% Change - Actual Cost	
January	121,398	121,221	-0.1%	947,708	1,327,490	+40.1%	+56.0%
February	74,797	146,623	+96.0%	632,778	1,450,103	+129.2%	-52.6%
March	84,131	91,040	+8.2%	711,751	900,382	+26.5%	-5.4%
April	77,702	85,743	+10.3%	652,357	858,756	+31.6%	-28.6%
May	71,294	82,704	+16.0%	578,906	992,447	+71.4%	-26.8%
June	62,104	77,044	+24.1%	504,285	924,531	+83.3%	-1.2%
July	71,398	44,991	-37.0%	579,753	539,887	-6.9%	-0.8%
August	74,204	67,705	-8.8%	698,258	697,361	-0.1%	+0.4%
September	74,426	72,270	-2.9%	700,353	744,377	+6.3%	-8.8%
October	92,146	80,109	-13.1%	926,633	825,127	-11.0%	+14.0%
November	99,191	74,805	-24.6%	1,128,799	659,028	-41.6%	+22.0%
December	96,475	110,651	+14.7%	1,097,889	974,840	-11.2%	+11.3%
Total/Average	**999,267**	**1,054,905**	**+5.6%**	**9,159,469**	**10,894,328**	**+18.9%**	**-5.5%**

Important note: *Minimum Outdoor Air Temperature (OAT) is the main variable taken into consideration on LPG consumption normalization.*

Energenz summarized the key items which could result in LPG consumption increase below:

- LPG Boiler consumption increased by 22.5% in December 2022 compared with the same month in 2021 as the main boiler has operated in the whole month of December 2022.

Table 13 presents the comparison of LPG rates in the years 2021 and 2022. The LPG rate in December 2022 decreased by **22.6%** compared with the same month in 2021.

Table 13. Macau - Summary of Actual Monthly LPG Rate

Monthly Period	Year 2021 – LPG Rate (MOP/kg)	Year 2022 – LPG Rate (MOP/ kg)	% Change – LPG Rate
January	7.81	10.95	+40.3%
February	8.46	9.89	+16.9%
March	8.46	9.89	+16.9%
April	8.40	10.02	+19.3%
May	8.12	12.00	+47.8%
June	8.12	12.00	+47.8%
July	8.12	12.00	+47.8%
August	9.41	10.30	+9.5%
September	9.41	10.30	+9.5%
October	10.06	10.30	+2.4%
November	11.38	8.81	-22.6%
December	11.38	8.81	-22.6%
Total/Average	**9.09**	**10.44**	**+14.8%**

Table 14 summarizes the monthly normalized LPG consumption. The actual LPG consumption in December 2022 increased by **14.0%** compared to 2019 normalized consumption in the same month.

Table 14. Macau - Summary of Normalized Monthly LPG Consumption

Monthly Period	2019 Normalized LPG Consumption (kg)	2022 Actual LPG Consumption (kg)	% Change – Normalized
January	95,134	121,221	+27.4%
February	117,728	146,623	+24.5%
March	78,544	91,040	+15.9%
April	85,634	85,743	+0.1%
May	70,235	82,704	+17.8%
June	72,880	77,044	+5.7%
July	66,446	44,991	-32.3%
August	64,919	67,705	+4.3%
September	76,029	72,270	-4.9%
October	71,468	80,109	+12.1%
November	73,184	74,805	+2.2%
December	97,043	110,651	+14.0%
Total/Average	**969,245**	**1,054,905**	**+8.8%**

2.3 ISO50001 Energy Consumption Record

The Macau ISO50001 energy performance target is to achieve 13% normalized energy savings by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to December 2022 is **10.8%** as shown in Table 15. Table 16 below summarizes the site monthly energy consumption (Electricity and LPG combined) compared to the 2019 baseline.

Table 15. Macau ISO 50001 Normalized Energy Performance Targets for 2025

MGM Properties	Baseline	Energy Savings	2021	Normalized Savings Achieved Up to December 2022	2023	2024	2025
Macau	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-5.9%**	**-10.8%**	-	-	**-**

Table 16. Macau ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2022 Actual Energy Consumption (GJ)	% Change – Normalized vs Actual
January	23,023	22,715	-1.3%
February	23,041	22,000	-4.5%
March	22,946	22,672	-1.2%
April	25,252	22,358	-11.5%
May	25,652	23,470	-8.5%
June	27,685	24,287	-12.3%
July	28,115	21,769	-22.6%
August	28,117	24,447	-13.1%
September	27,703	23,647	-14.6%
October	26,325	22,329	-15.2%
November	24,556	20,708	-15.7%
December	22,143	21,193	-4.3%
Total/Average	**304,557**	**271,596**	**-10.8%**

Important note: *Factors used for fuels' unit conversion: 1) Electricity (0.0036) and LPG (0.0453)*

2.4 Water Consumption

Based on the existing water metering system, the overall site water consumption can be broken down into four (4) categories: Kitchen and Restaurants; Hotel/Tower; Fire Services; and Chiller Plant & miscellaneous[2].

According to Table 17, the actual water consumption and cost in December 2022 both decreased by **13.4%** compared to the same month in 2021. To date, MGM Macau has decreased the actual water consumption and associated cost by **8.4%** and **8.3%** respectively.

Table 17. Macau - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2021	2022	% Change Actual	2021	2022	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	39,952	37,078	-7.2%	299,330	287,500	-4.0%	+15.8%	+36.7%
February	41,079	41,524	+1.1%	317,780	322,270	+1.4%	-90.8%	+43.2%
March	43,168	39,962	-7.4%	334,000	310,160	-7.1%	-6.9%	-9.6%
April	49,819	45,295	-9.1%	386,600	351,540	-9.1%	-11.4%	-11.4%
May	53,145	49,042	-7.7%	420,070	380,590	-9.4%	-32.6%	-12.2%
June	44,508	40,383	-9.3%	345,320	313,390	-9.2%	-1.0%	-35.0%
July	46,227	31,179	-32.6%	358,640	242,100	-32.5%	+4.1%	-92.5%
August	42,681	41,073	-3.8%	331,220	318,730	-3.8%	+1.7%	-17.7%
September	48,499	46,713	-3.7%	376,290	362,350	-3.7%	-0.1%	-17.4%
October	38,411	40,566	+5.6%	298,130	314,790	+5.6%	+7.9%	+0.9%
November	45,368	40,744	-10.2%	352,080	316,260	-10.2%	+14.7%	-21.8%
December	41,249	35,721	-13.4%	320,120	277,280	-13.4%	-93.6%	-37.3%
Total/Average	**534,106**	**489,280**	**-8.4%**	**4,139,580**	**3,796,960**	**-8.3%**	**-7.5%**	**-17.4%**

Important note: *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

Energenz summarized the key items which could result in water consumption decrease below:

- Hotel Tower water consumption for December 2022 has decreased by 9.4% compared to the same month in 2021.
- Podium water consumption for December 2022 has decreased by 6.6% compared to the same month in 2021.

[2] Miscellaneous includes the water consumption from toilets, laundry, etc.

Table 18 presents the comparison of water rates in the years 2021 and 2022. The water rate in December 2022 has increased by **0.02%** compared to the same month in 2021. The average utility rate of water has increased by **0.19%** compared to 2021.

Table 18. Macau - Summary of Actual Monthly Water Rates

Monthly Period	Year 2021 – Water Rate (MOP/m³)	Year 2022 – Water Rate (MOP/m³)	% Change – Water Rate
January	7.49	7.75	+3.5%
February	7.74	7.76	+0.3%
March	7.74	7.76	+0.3%
April	7.760	7.761	+0.01%
May	7.90	7.76	-1.8%
June	7.759	7.76	+0.02%
July	7.758	7.765	+0.09%
August	7.76	7.76	0%
September	7.759	7.757	-0.02%
October	7.762	7.76	-0.02%
November	7.761	7.762	+0.02%
December	7.76	7.76	+0.02%
Total/Average	**7.75**	**7.76**	**+0.19%**

Table 19 summarized the monthly normalized water consumption. The actual water consumption in December 2022 increased by **18.2%** compared to 2019 normalized consumption in the same month.

Table 19. Macau - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m³)	2022 Actual Water Consumption (m³)	% Change - Normalized
January	35,691	37,078	+3.9%
February	37,518	41,524	+10.7%
March	37,955	39,962	+5.3%
April	39,506	45,295	+14.7%
May	42,691	49,042	+14.9%
June	41,208	40,383	-2.0%
July	36,632	31,179	-14.9%
August	43,294	41,073	-5.1%
September	46,071	46,713	+1.4%
October	43,155	40,566	-6.0%
November	38,658	40,744	+5.4%
December	30,222	35,721	+18.2%
Total/Average	**472,601**	**489,280**	**+3.5%**

2.5 Macau Year 2022 Projects

Energenz has performed analysis and coordinated with MGM to implement the following projects in 2022 and reduced the site energy and water consumption which supports Energenz annual normalized energy savings target.

Table 20. MGM Macau Year 2022 Projects

Item	Initiatives	Status
Electricity & LPG		
1	Chilled Water Pump Retrofit (Phase 1)	Postpone to 2023
2	PICCV Implementation (Phase 3)	Completed
3	Air Side Equipment IE4 Motor Replacement	Completed
4	Neon Light LED Replacement (Phase 2)	Target to complete by early 2023
5	Replacement of L2M Steam-to-Water Heat Exchanger to Fully Utilize Heat Pump	Completed

3. MGM Cotai Performance

3.1 Electricity Consumption

The actual electricity consumption in December 2022 decreased by **8.8%** and the cost decreased by **5.9%** compared to the same month in 2021. To date, MGM Cotai has decreased actual electricity consumption by **9.9%** and associated costs decreased by **6.1%**

Table 21. Cotai - Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2021	2022	% Change - Actual	2021	2022	% Change - Actual Cost	
January	8,168,400	8,336,400	+2.1%	9,315,910	10,013,460	+7.5%	+15.8%
February	7,637,640	7,528,560	-1.4%	8,737,740	9,066,060	+3.8%	-90.8%
March	9,197,160	8,833,200	-4.0%	10,499,490	10,632,040	+1.3%	-6.9%
April	9,517,560	8,985,360	-5.6%	11,430,660	11,262,370	-1.5%	-11.4%
May	11,446,440	9,585,000	-16.3%	13,735,880	11,983,070	-12.8%	-32.6%
June	11,122,320	10,087,080	-9.3%	13,340,650	12,615,080	-5.4%	-1.0%
July	11,853,120	9,082,680	-23.4%	14,194,900	11,346,140	-20.1%	+4.1%
August	11,412,960	9,847,320	-13.7%	13,675,130	12,318,540	-9.9%	+1.7%
September	11,220,240	9,587,280	-14.6%	13,451,090	12,002,380	-10.8%	-0.1%
October	10,204,200	9,119,160	-10.6%	12,250,640	11,330,350	-7.5%	+7.9%
November	8,972,640	8,706,120	-3.0%	10,801,020	10,825,450	+0.2%	+14.7%
December	8,635,560	7,875,120	-8.8%	10,381,600	9,769,740	-5.9%	-93.6%
Total/Average	**119,388,240**	**107,573,280**	**-9.9%**	**141,814,710**	**133,164,680**	**-6.1%**	**-7.5%**

***Important note**: Cooling Degree Days (CDDs) is the main variable taken into consideration on electricity consumption normalization. The e-Bus electricity consumption has been included with the presented electricity consumption above.*

Energenz summarized the key items which could result in electricity consumption decrease below:

- Chiller Plant consumption for December 2022 has decreased by 35.4% compared to the same month in 2021 due to significant drop of CDD.
- Airside Equipment consumption for December 2022 has decreased by 8.0% compared to the same month in 2021.
- Kitchen and Restaurant consumption for December 2022 decreased by 15.6% compared to the same month in 2021.

Table 22 presents the monthly e-bus electricity consumption in MGM Cotai. The e-bus electricity consumption in December 2022 decreased by **1.8%** compared to the same month in 2021

Table 22. Cotai - Summary of Monthly e-bus Electricity Consumption

Monthly Period	E-bus Electricity Consumption (kWh)		
	2021	2022	% Change - Actual
January	44,839	42,287	-5.7%
February	32,534	47,905	+47.2%
March	34,902	38,793	+11.1%
April	37,412	38,537	+3.0%
May	51,584	51,328	-0.5%
June	56,629	36,420	-35.7%
July	64,410	0	-100.0%
August	44,185	31,160	-29.5%
September	40,618	35,281	-13.1%
October	30,528	36,433	+19.3%
November	41,515	27,466	-33.8%
December	36,070	35,422	-1.8%
Total/Average	**515,225**	**421,031**	**-18.3%**

Table 23 presents the comparison of electricity rates in the years 2021 and 2022. The electricity rate in December 2022 increased by **3.2%** compared to the same month in 2021.

Table 23. Cotai - Summary of Actual Monthly Electricity Rates

Monthly Period	Year 2021 – Electricity Rate (MOP/kWh)	Year 2022 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.14	1.20	+5.3%
February	1.14	1.20	+5.3%
March	1.14	1.20	+5.4%
April	1.20	1.25	+4.4%
May	1.20	1.25	+4.2%
June	1.20	1.25	+4.3%
July	1.20	1.25	+4.3%
August	1.20	1.25	+4.4%
September	1.20	1.25	+4.4%
October	1.20	1.24	+3.5%
November	1.20	1.24	+3.3%
December	1.20	1.24	+3.2%
Total/Average	**1.19**	**1.24**	**+4.3%**

Table 24 summarized the monthly normalized electricity consumption. The actual electricity consumption in December 2022 decreased by **5.8%** compared to 2019 normalized consumption in the same month.

Table 24. Cotai - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2022 Actual Electricity Consumption (kWh)	% Change - Normalized
January	8,635,397	8,336,400	-3.5%
February	8,394,419	7,528,560	-10.3%
March	9,473,790	8,833,200	-6.8%
April	9,911,574	8,985,360	-9.3%
May	10,366,415	9,585,000	-7.5%
June	11,183,378	10,087,080	-9.8%
July	11,776,857	9,082,680	-22.9%
August	11,425,668	9,847,320	-13.8%
September	11,583,550	9,587,280	-17.2%
October	10,647,628	9,119,160	-14.4%
November	9,828,041	8,706,120	-11.4%
December	8,363,804	7,875,120	-5.8%
Total/Average	**121,590,521**	**107,573,280**	**-11.5%**

3.2 Natural Gas Consumption

The actual natural gas consumption and cost in December 2022 decreased by **5.8%** and **1.1%** compared to the same month in 2021, respectively. To date, MGM Cotai has decreased actual natural gas consumption and associated costs by **32.2%** and **28.4%** respectively.

Table 25. Cotai - Summary of Actual Monthly Natural Gas Consumption and Cost

Monthly Period	Actual Natural Gas Consumption (m^3)			Actual Natural Gas Cost (MOP)			% Change – Min OAT	% Change – Food Covers
	2021	2022	% Change - Actual	2021	2022	% Change - Actual Cost		
January	60,643	59,378	-2.1%	363,990	376,240	+3.4%	+56.0%	+22.3%
February	54,221	76,089	+40.3%	325,930	482,130	+47.9%	-52.6%	+31.6%
March	68,066	52,146	-23.4%	407,980	330,410	-19.0%	-5.4%	-34.3%
April	86,866	43,900	-49.5%	518,960	278,170	-46.4%	-28.6%	-37.5%
May	90,102	45,632	-49.4%	537,940	289,140	-46.3%	-26.8%	-48.1%
June	80,606	33,949	-57.9%	482,270	215,120	-55.4%	-1.2%	-55.1%
July	102,473	3,041	-97.0%	610,450	19,270	-96.8%	-0.8%	-92.5%
August	51,837	37,574	-27.5%	311,790	238,080	-23.6%	+0.4%	-34.6%
September	51,460	48,351	-6.0%	309,560	306,370	-1.0%	-8.8%	-37.8%
October	49,326	49,413	+0.2%	296,870	313,100	+5.5%	+14.0%	+0.2%
November	57,386	43,924	-23.5%	344,680	278,320	-19.3%	+22.0%	-51.7%
December	63,501	59,825	-5.8%	383,112	379,080	-1.1%	+11.3%	-49.9%
Total/Average	**816,487**	**553,222**	**-32.2%**	**4,893,532**	**3,505,430**	**-28.4%**	**-5.5%**	-35.9%

Important note: *Minimum Outdoor Air Temperature (OAT is the main variable taken into consideration on natural gas consumption normalization.*

Energenz summarized the key items which could result in natural gas consumption decrease below:

- Kitchen NG consumption for December 2022 has decreased by 25.7% and food cover has decreased 49.9% compared to the same month in 2021.

Table 26 presents the comparison of natural gas rates in the years 2021 and 2022. The natural gas rate in December 2022 increased by **5.0%** compared with the same month in 2021.

Table 26. Cotai - Summary of Actual Monthly Natural Gas Rates

Monthly Period	Year 2021 – Natural Gas Rate (MOP/m³)	Year 2022 – Natural Gas Rate (MOP/m³)	% Change – Natural Gas Rate
January	6.00	6.34	+5.6%
February	6.01	6.34	+5.4%
March	5.99	6.34	+5.7%
April	5.97	6.34	+6.1%
May	5.97	6.34	+6.1%
June	5.98	6.34	+5.9%
July	5.96	6.34	+6.4%
August	6.01	6.34	+5.3%
September	6.02	6.34	+5.3%
October	6.02	6.34	+5.3%
November	6.01	6.34	+5.5%
December	6.03	6.34	+5.0%
Total/Average	**6.00**	**6.34**	**+5.6%**

Table 27 summarized the monthly normalized natural gas consumption. The actual natural gas consumption in December 2022 decreased by **52.9%** compared to 2019 normalized consumption in the same month. The normalized model and comparison of natural gas has been modified and included since January 2022.

Table 27. Cotai - Summary of Normalized Monthly Natural Gas Consumption

Monthly Period	2019 Normalized Natural Gas Consumption (m³)	2022 Actual Natural Gas Consumption (m³)	% Change - Normalized
January	125,664	59,378	-52.7%
February	132,723	76,089	-42.7%
March	119,389	52,146	-56.3%
April	122,527	43,900	-64.2%
May	116,565	45,632	-60.9%
June	105,271	33,949	-67.8%
July	103,702	3,041	-97.1%
August	103,859	37,574	-63.8%
September	104,957	48,351	-53.9%
October	113,428	49,413	-56.4%
November	115,310	43,924	-61.9%
December	127,076	59,825	-52.9%
Total/Average	**1,390,471**	**553,222**	**-60.2%**

3.3 ISO50001 Energy Consumption Record

Cotai ISO50001 energy target is to achieve 13% normalized energy consumption reduction by 2025 compared to the 2019 baseline. Based on the official agreement with MGM Cotai, e-bus consumption should be included in the energy consumption tracking as part of the ISO50001 requirement. To date, normalized energy saving achieved up to December 2022 is **16.9%** as shown in Table 28 and Table 29 below summarizes the site monthly energy consumption (Electricity and Natural Gas combined) compared to the 2019 baseline.

Table 28. Cotai ISO 50001 Normalized Energy Performance Targets for 2025

MGM Properties	Baseline	Energy Savings	2021	Normalized Savings Achieved Up to December 2022	2023	2024	2025
Cotai	2019	Target	-4%	-8%	-12%	N/A	**-13%**
		Actual	**-6.4%**	**-16.9%**	-	-	-

Table 29. Cotai ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2022 Actual Energy Consumption (GJ)	% Change - Normalized vs Actual
January	35,567	32,071	-9.8%
February	36,708	29,743	-19.0%
March	37,322	33,609	-9.9%
April	40,715	33,871	-16.8%
May	41,466	36,089	-13.0%
June	44,419	37,492	-15.6%
July	45,330	32,803	-27.6%
August	45,073	36,754	-18.5%
September	44,441	36,192	-18.6%
October	42,995	34,544	-19.7%
November	39,614	32,866	-17.0%
December	35,653	30,426	-14.7%
Total/Average	**489,303**	**406,461**	**-16.9%**

***Important note**: Factors used for fuels' unit conversion: 1) Electricity (0.0036) and Natural Gas (0.0347), e-bus consumption is included*

3.4 Water Consumption

According to Table 30, the actual water consumption and associated costs in December 2022 both decreased by **29.7%** compared to the same month in 2021. To date, MGM Cotai has decreased actual water consumption and associated costs by **21.6%** and **21.7%** respectively.

Table 30. Cotai - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2021	2022	% Change Actual	2021	2022	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	43,292	44,658	+3.2%	336,700	347,240	+3.1%	+15.8%	+22.3%
February	50,837	45,298	-10.9%	395,180	352,330	-10.8%	-90.8%	+31.6%
March	55,668	44,914	-19.3%	432,400	349,260	-19.2%	-6.9%	-34.3%
April	65,511	51,640	-21.2%	509,000	401,500	-21.1%	-11.4%	-37.5%
May	77,506	55,628	-28.2%	612,200	432,480	-29.4%	-32.6%	-48.1%
June	66,115	45,805	-30.7%	513,500	356,090	-30.7%	-1.0%	-55.1%
July	70,716	38,031	-46.2%	549,160	295,890	-46.1%	+4.1%	-95.5%
August	62,738	49,076	-21.8%	487,380	381,480	-21.7%	+1.7%	-34.6%
September	70,532	55,797	-20.9%	547,710	433,480	-20.9%	-0.1%	-37.8%
October	55,439	51,216	-7.6%	430,760	398,090	-7.6%	+7.9%	+0.2%
November	57,380	52,513	-8.5%	445,950	408,250	-8.5%	+14.7%	-51.7%
December	58,227	40,909	-29.7%	452,410	318,230	-29.7%	-93.6%	-49.9%
Total/ Average	**733,961**	**575,485**	**-21.6%**	**5,712,350**	**4,474,320**	**-21.7%**	**-7.5%**	**-35.9%**

***Important note**: Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

Energenz summarized the key items which could result in water consumption decrease below:

- Hotel Tower water consumption for December 2022 has decreased by 26.2% and occupancy has decreased by 33.0% when compared with the same month last year.
- CT Make Up consumption for December 2022 has decreased by 29.4% when compared with the same month last year.

Table 31 presents the comparison of water rates in the years 2021 and 2022. The water rate in December 2022 increased by **0.1%** compared to the same month in 2021. The average utility rate of water has decreased by **0.07%** compared to 2021.

Table 31. Cotai - Summary of Actual Monthly Water Rates

Monthly Period	Year 2021 – Water Rate (MOP/m³)	Year 2022 – Water Rate (MOP/m³)	% Change – Water Rate
January	7.777	7.776	-0.02%
February	7.77	7.78	+0.1%
March	7.77	7.78	+0.1%
April	7.770	7.775	+0.1%
May	7.90	7.77	-1.6%
June	7.767	7.774	+0.1%
July	7.77	7.78	+0.2%
August	7.768	7.773	+0.1%
September	7.765	7.769	+0.04%
October	7.770	7.773	+0.04%
November	7.772	7.774	+0.03%
December	7.77	7.78	+0.1%
Total/Average	**7.78**	**7.775**	**-0.07%**

Table 32 summarized the monthly normalized water consumption. The actual water consumption in December 2022 increased by **0.9%** compared to 2019 normalized consumption in the same month.

Table 32. Cotai - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m³)	2022 Actual Water Consumption (m³)	% Change - Normalized
January	48,361	44,658	-7.7%
February	49,015	45,298	-7.6%
March	49,249	44,914	-8.8%
April	51,413	51,640	+0.4%
May	54,331	55,628	+2.4%
June	56,037	45,805	-18.3%
July	54,785	38,031	-30.6%
August	59,454	49,076	-17.5%
September	61,158	55,797	-8.8%
October	59,726	51,216	-14.2%
November	48,428	52,513	+8.4%
December	40,542	40,909	+0.9%
Total/Average	**632,498**	**575,485**	**-9.0%**

3.5 Cotai Year 2022 Projects

Energenz has performed analysis and coordinated with MGM to implement the following projects in 2022 and reduced the site energy and water consumption which supports Energenz annual normalized energy savings target.

Table 33. MGM Cotai Year 2022 Projects

Item	Initiatives	Status
Electricity & NG		
1	Chiller Plant Control Optimization (Chiller/CHWP/CDWP/CT)	On-going
2	Hot Water System Upgrade 1. Increase Max. No of Plate for Pre-heat Compact Unit 2. Upgrade Booster Compact Unit 3. Install Electric Heater for Booster Calorifier	On-going
3	AHU Static Pressure & Supply Air Temperature Set-Point Reset (Phase 2)	Postpone to 2023
4	PICCV Implementation (Phase 3)	Completed
5	Hotel Towers Building Pressurization and EAF Optimization	Completed
6	Kitchen Exhaust Fan (KEF) Steps Speed Control Optimization	Completed
7	Carpark Ventilation Fans Control Optimization with IoT Sensor Installation (Phase 2)	Completed
8	Centralized RO Pump Set Replacement (10 Units)	Target to complete by mid 2023
Water		
9	Extension of CDP Water Recycling (Phase 3 & 4)	Completed
10	Rainwater Harvesting for CT Make Up (Phase 1)	Completed
11	Non-potable Centralized RO Optimization for Cooling Tower	Completed

Appendix A

Regression Analysis

The 2019 normalized consumption modelling is remained used for tracking the agreed normalized energy savings target of 4% accurately for MGM Macau and MGM Cotai. The following tables show a summary of the selected independent variables for each utility which are used for the normalized consumption modelling in 2022.

MGM Macau

Utilities	2022	
Electricity (kWh)	CDD (Base T=17°C)	
LPG (kg)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

MGM Cotai

Utilities	2022	
Electricity (kWh)	CDD (Base T=17°C)	
Natural Gas (m^3)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

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